

PRACTICAL-14PACKET SNIFFINGAIM:-

To implement packet sniffing using RAW sockets.

CODE:-

```
from scapy.all import sniff
from scapy.layers.inet import IP, TCP, UDP, ICMP

def packet_callback(packet):
    if IP in packet:
        ip_layer = packet[IP]
        protocol = ip_layer.proto
        src_ip = ip_layer.src
        dst_ip = ip_layer.dst
        protocol_name = ""
        if protocol == 1:
            protocol_name = "ICMP"
        elif protocol == 6:
            protocol_name = "TCP"
        elif protocol == 17:
            protocol_name = "UDP"
        else:
            protocol_name = "Unknown Protocol"
        print(f"Protocol : {protocol_name}")
        print(f"Source IP: {src_ip}")
        print(f"Destination IP: {dst_ip}")
        print("-" * 50)

sniff(
    iface = "Wi-Fi",
    prn = packet_callback,
    filter = "ip",
    store = 0
)
```

OUTPUT:-

python practical14-sniff.py

Protocol : UDP

Source IP: 104.18.41.41

Destination IP: 10.184.154.113

Protocol: UDP

Source IP: 10.184.154.113

Destination IP: 104.18.41.41

Protocol: UDP

Source IP: 10.184.154.113

Destination IP: 10.184.154.24

Protocol: UDP

Source IP: 10.184.154.113

Destination IP: 10.184.154.24

Protocol: UDP

Source IP: 10.184.154.24

Destination IP: 10.184.154.113

Protocol: UDP

Source IP: 10.184.154.24

Destination IP: 10.184.154.113

Protocol: UDP

Source IP: 10.184.154.113

Destination IP: 10.184.154.24

Protocol: TCP

Source IP: 10.184.154.113

Destination IP: 104.46.162.231

### RESULT :-

This packet sniffer using raw sockets is implemented successfully.